Course Name: Introduction to Software Engineering

Course Code: IS1103-1

Assignment Number & Title: System Modeling and Agile Software Development – Bharat Taxi System

Student Name: Aishwarya.k.shetty

USN / Roll Number: NNM24IS012

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Department: Information Science & Engineering

Institution Name: NMAM Institute of Technology (NMAMIT)

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Facutly Name: Dr. Jason Elroy Martis

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Abstract

Ride booking applications are becoming very common in daily life. Many people use mobile apps to book taxis, autos, or bikes for travelling. Bharat Taxi is one such service that connects passengers with drivers using a mobile application. The system helps users to book rides, find nearby drivers, track the trip, and complete payment after the ride.

In this assignment, the Bharat Taxi system is taken as an example to understand system modeling and Agile development concepts. Different models such as context model, interaction model, and behavioral model are used to explain how the system works internally.

The report also explains how Agile development can be used while building such applications. Scrum framework, sprint planning, user narrative, and definition of done are discussed. These concepts help developers build the system step by step and improve it based on feedback.

Github repository:

Introduction

Nowadays transportation apps have become part of our everyday life. Instead of waiting for taxis on the road, people can easily book a ride using a mobile phone. Applications like Bharat Taxi make travelling easier by connecting passengers with nearby drivers.

The Bharat Taxi platform works through a mobile application where passengers can request rides and drivers can accept those requests. The system checks available drivers, assigns a driver to the passenger, and tracks the ride using GPS.

In software engineering, system modeling is used to understand how such systems are designed and how different parts of the system interact with each other. In this assignment, system models are used to represent the working of the Bharat Taxi system. Agile development practices are also discussed to understand how the system can be developed step by step.

Problem Statement

The main goal of this assignment is to design a simple system model for a ride-hailing application similar to Bharat Taxi. The system should allow passengers to request rides and drivers to accept those requests using the mobile application.

The system should also manage other functions like driver assignment, ride tracking, and payment after the ride. In addition to system modeling, Agile development using Scrum methodology is also explained to show how the application can be developed in stages.

System Modeling

System modeling is used to represent the structure and behavior of a system using diagrams. These diagrams help in understanding how different components of the system communicate with each other.

For the Bharat Taxi system, modeling helps us see how passengers, drivers, and the system interact during the ride process. In this assignment three models are mainly used:

- Context Model
- Interaction Model
- Behavioral Model

These models help explain the overall working of the Bharat Taxi platform.

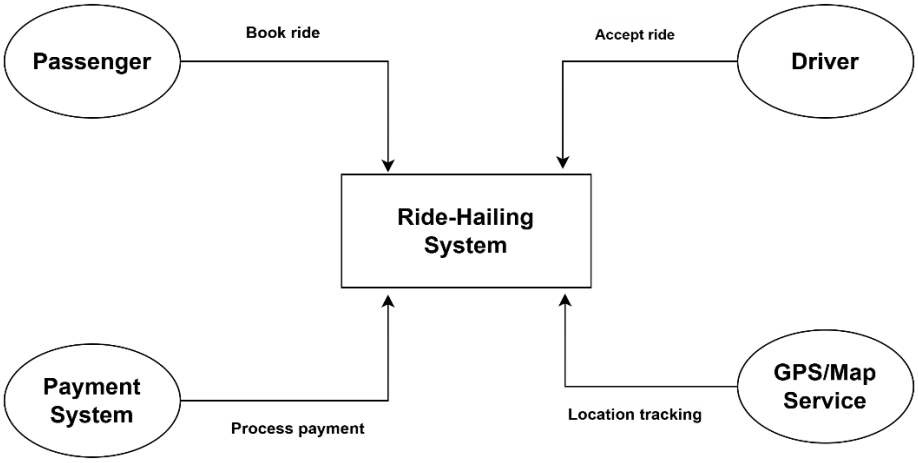
- ## Context Model

The context model shows the Bharat Taxi system and the external entities connected to it. In this system, the main external entities are passengers, drivers, GPS services, payment gateway, and system administrator.

Passengers use the application to book rides by entering their pickup and destination locations. The system then checks for drivers who are available nearby and sends the request to them. Drivers receive the ride request through their application and can accept or reject the request.

GPS services help track the location of the driver and passenger during the trip. After the ride is finished, the payment gateway processes the payment. The

system administrator manages the platform and monitors the activities in the system.

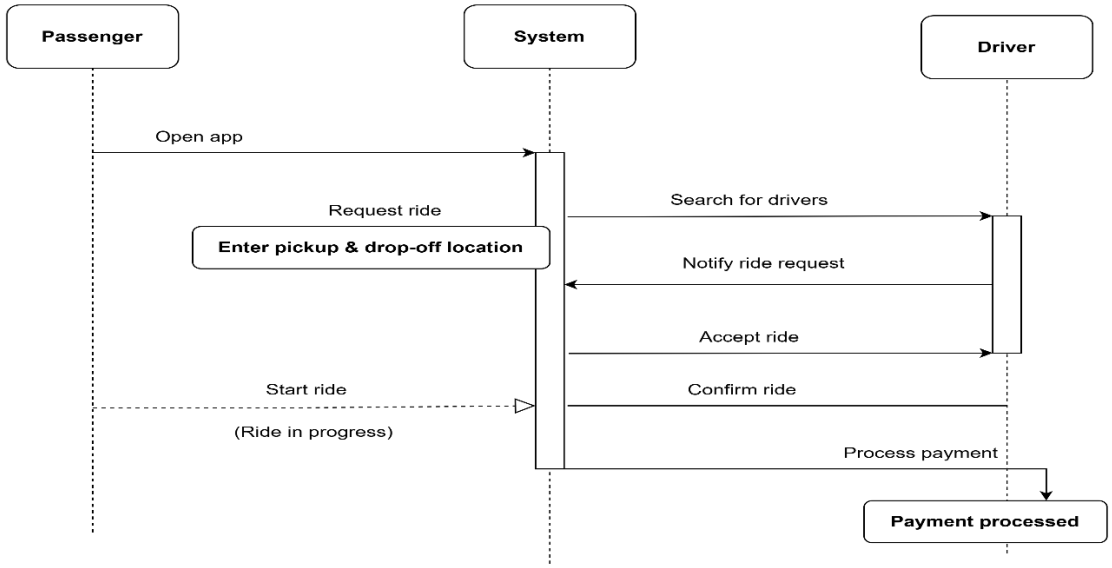


- ## Interaction Model (Sequence Diagram)

The interaction model explains the communication between the passenger, driver, and the system while booking a ride.

First, the passenger opens the Bharat Taxi app and enters the ride details. The system receives this request and searches for drivers who are available nearby. The ride request is then sent to one or more drivers.

Once a driver accepts the request, the system confirms the booking and shows the driver details to the passenger. The driver travels to the pickup location and starts the ride. After reaching the destination, the ride is completed and the payment process takes places.



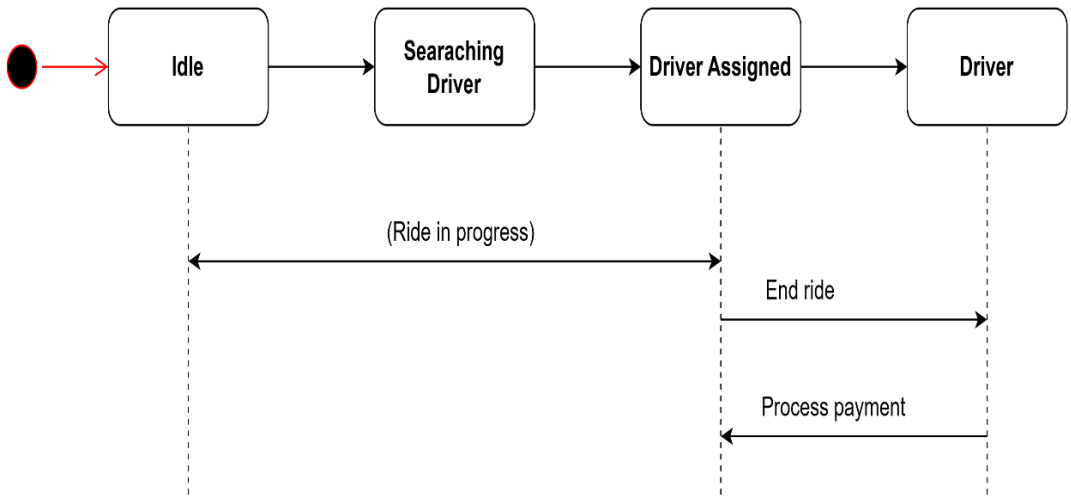
- **Behavioral Model (State Diagram)**

The behavioral model describes the different stages that happen during a ride.

At the beginning the system is in an idle state. When a passenger requests a ride, the system moves to the ride request state. The system then looks for available drivers and assigns one driver to the passenger.

After the driver accepts the request and arrives at the pickup location, the ride starts. During the journey the system stays in the ride in progress state. When the passenger reaches the destination, the ride is marked as completed and payment is processed.

Finally the system goes back to the idle state and waits for the next ride request.



Agile Method

Agile is a software development approach where the system is developed step by step instead of building everything at once. The work is divided into smaller parts and each part is completed in a short period of time.

This method helps developers test features quickly and make improvements whenever required. For applications like Bharat Taxi, Agile development makes it easier to add new features and update the system regularly.

- **Overview of Scrum**

Scrum is one of the most popular frameworks used in Agile development. In Scrum, the work is divided into small development cycles called sprints. Each sprint usually lasts for one or two weeks.

During every sprint the development team works on a few selected tasks. Scrum also defines roles such as Product Owner, Scrum Master, and Development Team. These roles help in organizing the development process and completing the project efficiently.

- **Sprint Planning**

Sprint planning is a meeting where the development team decides what tasks should be completed during the next sprint. The team selects tasks from the project backlog and discusses how they will complete them.

This helps the team understand their work clearly and ensures that everyone knows their responsibilities for that sprint.

- **User Narrative**

A user narrative describes a feature of the system from the user's point of view. It explains what the user wants to do using the system and why that feature is useful.

User narratives help developers understand the needs of passengers and drivers so that the system can be designed in a way that improves user experience.

- **Definition of Done**

The definition of done explains when a task or feature can be considered complete. It includes certain conditions that must be satisfied before the feature is delivered.

This ensures that the feature has been properly implemented, tested, and works correctly before it becomes part of the final system.

Examples of User Narrative with Definition of Done

➢ Passenger User Stories

❖ Example 1 – Passenger Registration

- User Story: As a passenger, I want to create an account so that I can use the Bharat Taxi app.
- Definition of Done: The task is done when the passenger enters basic details and the account is created.

❖ Example 2 – Passenger Login

- User Story: As a passenger, I want to log in to the app so that I can book a ride.
- Definition of Done: The task is done when the passenger enters correct details and the app opens.

❖ Example 3 – Book a Ride

- User Story: As a passenger, I want to enter pickup and destination so that I can request a ride.
- Definition of Done: The task is done when the system sends the request to drivers and a driver accepts it.

❖ Example 4 – Track Driver

- User Story: As a passenger, I want to see the driver location on the map so that I know when the driver will arrive.
- Definition of Done: The task is done when the passenger can see the driver location in the app.

❖ Example 5 – Make Payment

- User Story: As a passenger, I want to pay for the ride after reaching my destination.
- Definition of Done: The task is done when the payment is completed and the ride ends.

➢ Driver User Stories

❖ Example 1 – Driver Registration

- User Story: As a driver, I want to register in the app so that I can receive ride requests.
- Definition of Done: The task is done when the driver submits personal and vehicle details.

❖ Example 2 – Driver Login

- User Story: As a driver, I want to log in so that I can see ride requests.
- Definition of Done: The task is done when the driver logs in and the dashboard opens.

❖ Example 3 – Accept Ride

- User Story: As a driver, I want to accept ride requests so that I can pick up passengers.
- Definition of Done: The task is done when the driver accepts the request in the app.

❖ Example 4 – Go to Pickup Location

- User Story: As a driver, I want to see the passenger location so that I can reach there.
- Definition of Done: The task is done when the map shows the route to the passenger.

❖ Example 5 – Complete Ride

- User Story: As a driver, I want to end the ride after reaching the destination.
- Definition of Done: The task is done when the driver ends the ride and the fare is shown.

Discussion

While working on this assignment, it becomes clear that system modeling makes complex systems easier to understand. The context model gives a high level idea of the system, while the sequence diagram explains the interaction between different users.

The state diagram shows how the ride moves through different stages from booking to completion. Agile methods like Scrum also help developers organize their work and build the system step by step.

Conclusion

In this assignment, the Bharat Taxi ride-hailing system was studied using different system modeling techniques. Context model, sequence diagram, and state diagram helped in understanding how the system works.

Agile development concepts such as Scrum and sprint planning were also discussed. These methods help development teams build applications in an organized and flexible way.

Overall, system modeling and Agile development play an important role in designing modern applications like Bharat Taxi.

References

[1] Ian Sommerville, Software Engineering, Pearson

[2] Roger Pressman, Software Engineering: A Practitioner's Approach

[3] Scrum Guide – https://scrumguides.org

Declaration

I hereby declare that this assignment report is my original work prepared for academic purposes. All sources used in this report have been properly referenced.

Student signature:

Date : 03-04-2026